SOIL EROSION AND CONSERVATION · NOTE ER09

Runoff and Soil-Loss Measurement — Stream Gauging, Stage Recorders, Runoff Plots and Sediment Samplers

Course topics: SWCE3.5 L10, SWCE4.5 L6, SWCE-502 T3, SWCE-505 T6, SWCE-501 T11

Stream gauging

- Water-measuring devices also measure runoff; flumes and weirs are the commonest. Runoff measurement needs special devices, a record of water stage (head) and sampling of sediment load — together called **stream gauging**, i.e. measuring discharge at a channel section.
- Overland flow cannot be measured directly — no suitable instrument, and it varies too much with topography, vegetation, soil and rain. Channel flow includes overland flow plus subsurface flow intercepted by the stream; long after rain it is mainly subsurface flow, which is often negligible in small watersheds.
- **Streamflow data** are needed for soil-conservation and flood-control design, dependable low-flow supply, storage design (tanks, reservoirs) and understanding runoff variability.

Gauging site and control

- Natural sections are unreliable — irregular, changing with erosion and sedimentation, with unstable stage–discharge relations. Accuracy improves with a regular section, a weir or flume, and staff gauges or automatic recorders.
- **Stage (head):** height of the water surface above a datum; a stage–discharge curve converts stage to discharge. Records come from non-recording or recording gauges or electronic transmission.
- **Datum:** below the zero-flow level, preferably referred to mean sea level, permanent for the life of the station, and tied to independent bench marks.
- **Control:** a section where water level is a stable index of discharge — on a straight, stable reach with a stable head–discharge relation, considering velocity, slope, shape, roughness, absence of tail-water effect and enough depth at the lowest stage.
- **Site selection:** permanent control; no backwater; straight, uniform reach ≥ 3–5 stream widths upstream and ≥ 2 widths downstream; flow normal to the section; stable banks and bed; easy access (near a road).

Discharge measurement

- Small watersheds: pre-calibrated flumes or weirs. Large watersheds: the velocity–area method with a current meter, whose rotating cups or vanes give velocity from their revolutions; velocity is measured at several depths and widths and the area is surveyed.

$$Q = a \times v \text{ or, by sub-sections, } Q = \Sigma\, a_i \cdot v_i$$

$$a = \text{cross-sectional area } (m^2);\ v = \text{average velocity } (m/s)$$

Correction: the notes say "surface velocity ≈ 80–85% of average velocity". It is the other way round — the average velocity is about 80–85% of the surface velocity, because flow is fastest near the surface.

Stage gauges

Gauge	Description
Staff gauge	Vertical or inclined enamelled scale fixed firmly near the bank, zero below the lowest water level
Chain gauge	Horizontal scale, chain and weight on a bridge or structure; read when the weight touches the water
Wire-and-weight gauge	Reel-wound wire with a weight, usually in a stilling well; gives water-surface elevation above datum
Hook gauge	Most reliable non-recording type: the point is raised until it disturbs the surface; a vernier reads to a fraction of a mm. Rarely used in routine gauging
Recording gauges	Continuous stage record (analog chart or digital); give an accurate daily mean stage, maximum and minimum stages with times, data without an observer, and no human error

- **Graphical recorders:** a clock (spring, weight or motor driven) turns a vertical or horizontal drum once in 24 hours or 8 days while a float with a counterweight drives the pen.
- **Stilling well:** the recorder sits in a well beside the stream so the float follows the level without waves. Well depth > 2 × the expected stream depth + 15 cm (as printed); several intake pipes against clogging, each ≈ 1/10 of the well diameter; circular or rectangular; access pipe at least ≈ 60 cm; ventilation and maintenance access. Set the datum below zero flow and survey it in (preferably to mean sea level); check periodically for settlement, blocked pipes, clock operation and equal levels in well and stream.
- **Digital recorders:** punched-paper-tape recorders turn float motion into coded punches (a four-group code for 0–9999) for computer use; analog-to-digital data loggers take pressure, bubbler or angular transducer signals and suit satellite or microwave transmission.
- **Modern station:** an approach section with flume or weir, continuous stage recorder, data logger and transfer system, processing software and sediment sampler. Telemetry by telephone, wireless or satellite gives real-time stage and discharge for flood forecasting, reservoir regulation and online data.

Measuring soil loss: runoff plots

Soil-loss measurements show the effect of conservation practices and give relations for predicting loss under different soils and management — mainly from runoff plots.

- **Site:** slope typical of the area and uniform (avoid grading that disturbs the profile or exposes subsoil); near the farmstead; no trees within 3 × tree height; a natural drop downstream for runoff disposal; uniform soil in all plots. If grading is unavoidable, scrape off ~7.5 cm of topsoil, store it, and spread it back evenly afterwards.
- **Size:** from micro-plots to large plots — larger plots are more realistic but harder to manage. The standard for USLE studies is **2 m × 22 m**, separated by 22.5 cm brick walls in cement mortar or removable GI sheets, with a 60 cm path between plots. Plots need not be parallel; orient them to get the required grades.
- **Collection:** converging low walls lead runoff to a baffled stilling tank; the overflow enters a multi-slot divisor of uniform depth; a representative sample goes to the receiving tank and the rest is drained. The receiving tank should hold ≥ 12 hours of runoff from maximum rainfall; protect downstream slopes with a retaining wall.

Multi-slot divisor

- Measures runoff and sediment from plots by passing only a fraction of the flow to the collection tank — automatically and continuously. Types: the stationary multi-slot divisor and the Coshocton wheel. It still needs fairly large tanks.
- **Layout:** runoff → conveyance channel → sludge tank (heavier sediment settles) → multi-slot divisor → aliquot tanks.
- **Design:** size for the maximum runoff rate (taken as the maximum 5-minute rainfall rate) and sediment load; storage for 100% runoff of the 100-year, 48-hour storm; conveyance velocity ≥ ≈ 60 cm/s so sediment does not settle.
- **Details:** the collector acts as a weir at the plot end; its end plate goes ≥ 20 cm into the soil. The divisor is about as deep as the conveyance channel plus ~12 cm freeboard and 20–25 cm wide; 1.5–2.5 cm mesh screens keep out trash. Build in galvanised sheet steel and angles (stainless steel for precision plates); level it, connect it to the sample tanks with drains, and provide drainage and covers.

Coshocton wheel sampler

- Developed at Coshocton, USA; used with a multi-slot divisor or H-flume for runoff and sediment from small farm watersheds.
- A precision galvanised circular wheel with eight vanes (slot width checked with a taper gauge). Flow from the H-flume strikes the inclined wheel and turns it; an elevated sampling slot takes an aliquot each revolution and routes it to a storage tank.
- Commonly for watersheds ≤ 1 ha. A sediment sampler developed at ICRISAT, Hyderabad, works up to 400 ha but has not been widely field-tested.

Calibration: calibrate plots and divisors before a study, with uniform cropping and slope. Pour a known quantity of water upstream, measure the aliquot and record the division error — largest at the lowest flows — then choose the number and depth of slots accordingly.

Runoff measuring devices

- Prefabricated triangular weirs and **H-flumes**. An H-flume confines flow to a set section and length and works on the minimum-energy (critical-depth) principle without ponding water upstream as a weir does; head is read in a stilling well.
- Types HS, H and HL (NRCS, USA): HS for very small flows with high accuracy; H for 0.009–0.85 m^3/s (a 15 cm deep H-flume carries at most 0.009 m^3/s); HL for larger flows. Only 1–2 m flumes are recommended for agricultural watersheds.
- Build them precisely in galvanised sheet steel and angles; set them in an approach box below ground with an upstream concrete cut-off wall and watertight bolted joints, plus collection gutters where the slope is small.

Sediment load in large watersheds

- Sediment moves in suspension (suspended load) and by saltation and surface creep (bed load); total sediment = suspended + bed load.
- **Suspended-sediment samplers** (depth-integrating and point-integrating) must take water in at the local stream velocity, face upstream outside the disturbed zone, fill smoothly without gulping, sample near the bed, and be streamlined, heavy and rugged.

- **Depth-integrating sampler:** sampled continuously while lowered from surface to bed and back at constant speed; compressed air balances the hydrostatic pressure. Lowering too fast causes pressure error, too slowly over-filling — fill to about 2/3 to 3/4 of the container.
- **Point-integrating sampler:** an electrically controlled rotary valve samples one point for a short time.
- **Bed load:** sample from the stream bed or reservoir deposits; transport rate = mass collected per unit time ÷ sampling width.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA-NRCS, National Engineering Handbook Part 650 (H-flumes). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The standard runoff plot for USLE studies in India is

(a) 1 m × 10 m
(b) 2 m × 22 m
(c) 5 m × 50 m
(d) 10 m × 100 m

Q2. Runoff plots should be at least ___ tree heights from trees

(a) 1
(b) 2
(c) 3
(d) 10

Q3. Average velocity in a stream is about ___ of surface velocity

(a) 50%
(b) 65%
(c) 80–85%
(d) 100%

Q4. A gauging reach should be straight for at least ___ stream widths upstream

(a) 1
(b) 3–5
(c) 10
(d) 20

Q5. The Coshocton wheel has

(a) four vanes
(b) six vanes
(c) eight vanes
(d) twelve vanes

Q6. Multi-slot divisor storage is designed for 100% runoff of the

(a) 10-yr 24-h storm
(b) 25-yr 6-h storm
(c) 100-yr 48-h storm
(d) 1000-yr 1-h storm

Q7. Discharge range of an H-flume is about

(a) 0.001–0.01 m^3/s
(b) 0.009–0.85 m^3/s
(c) 1–10 m^3/s
(d) 10–100 m^3/s

Q8. A depth-integrating sampler should be filled to about

(a) 1/4
(b) 1/2
(c) 2/3 to 3/4
(d) completely

Q9. The most reliable non-recording stage gauge is the

(a) staff gauge
(b) chain gauge
(c) hook gauge
(d) wire-and-weight gauge

Answers

Q1 — (b) 2 m × 22 m
Q2 — (c) 3
Q3 — (c) 80–85%
Q4 — (b) 3–5

Q5 — (c) eight vanes
Q6 — (c) 100-yr 48-h storm
Q7 — (b) 0.009–0.85 m³/s
Q8 — (c) 2/3 to 3/4
Q9 — (c) hook gauge